

Chapter 3: Interdependence and the Gains from Trade

Chapter Objectives (1 of 2)

By the end of this chapter, you should be able to:

- Explain how the terms of trade can lead to gains.
- Describe opportunity cost in the context of the production possibilities frontier.
- Describe how the production possibilities frontier explains aggregate production.
- Determine if an output level is exhibiting allocative or productive efficiency.
- Explain how comparative advantage determines trade.
- Describe absolute advantage in the context of trade.

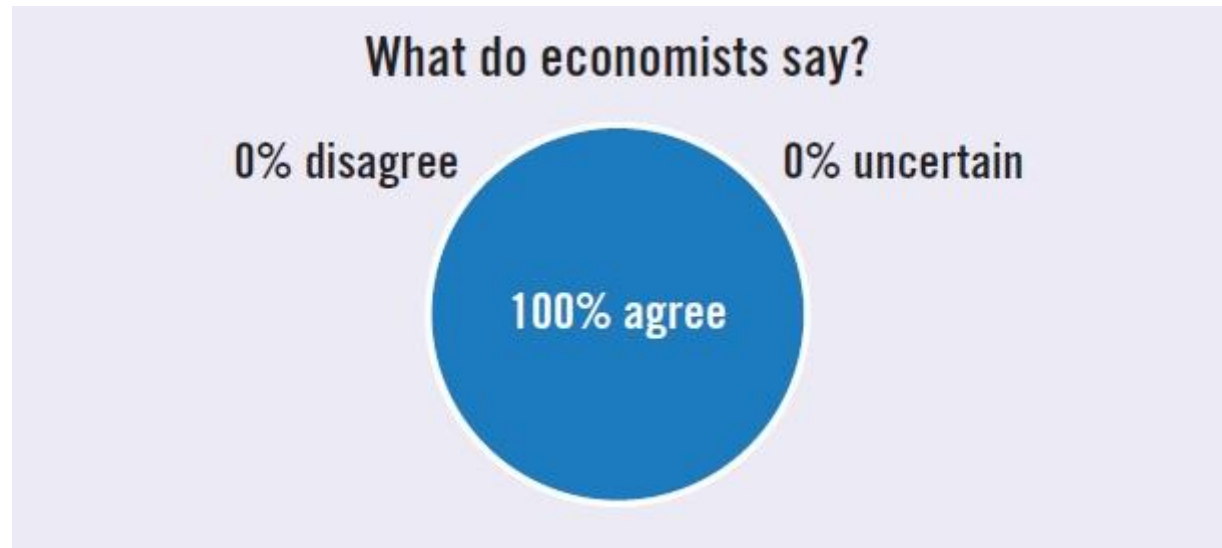
Chapter Objectives (2 of 2)

- Describe the factors that cause the production possibilities frontier to shift.
- Determine whether a country will be an importer or exporter of a good if the country opens up to international trade.

3.1 A Parable for the Modern Economy

Ask the Experts: Trade between China and the United States - A

“Trade with China makes most Americans better off because, among other advantages, they can buy goods that are made or assembled more cheaply in China.”



Source: IGM Economic Experts Panel, June 19, 2012.

Economic Interdependence

- “Trade can make everyone better off”
 - One of the Ten Principles of Economics
- We now examine this principle more closely
 - What exactly do people gain when they trade with one another?
 - Why do people become interdependent?

A Simple Economy

- Only two goods
 - Meat
 - Potatoes
- Only two people
 - A cattle rancher named Ruby
 - A potato farmer named Frank
- Both would like to eat both meat and potatoes

Production Possibilities

- If Ruby produces only meat and Frank produces only potatoes
 - Both gain from trade
- If both Ruby and Frank produce both meat and potatoes
 - Both gain from specialization and trade
- Production possibilities frontier
 - Various mixes of output that an economy can produce

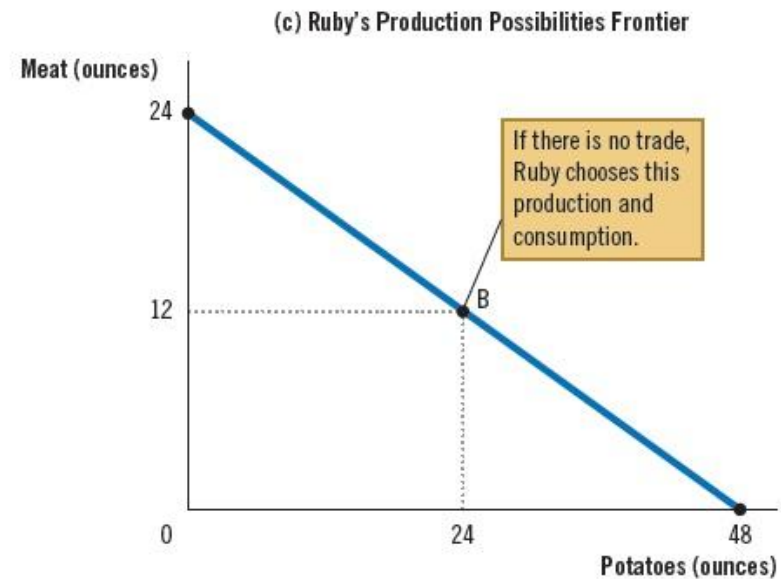
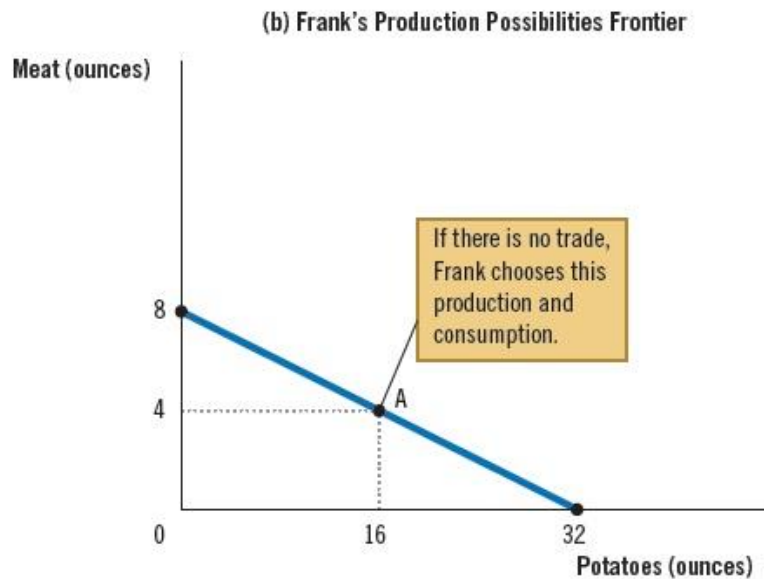
Figure 1 The Production Possibilities Frontier (1 of 2)

Panel (a) shows the production opportunities available to Frank the farmer and Ruby the rancher.

(a) Production Opportunities				
	Minutes Needed to Make 1 Ounce of:		Amount Produced in 8 Hours	
	Meat	Potatoes	Meat	Potatoes
Frank the farmer	60 min/oz	15 min/oz	8 oz	32 oz
Ruby the rancher	20 min/oz	10 min/oz	24 oz	48 oz

Figure 1 The Production Possibilities Frontier (2 of 2)

Panel (b) shows the combinations of meat and potatoes that Frank can produce. Panel (c) shows the combinations of meat and potatoes that Ruby can produce. Both production possibilities frontiers assume that Frank and Ruby each work 8 hours per day. If there is no trade, their production possibilities frontiers are also their consumption possibilities frontiers.



Example 1: The U.S. PPF (1 of 3)

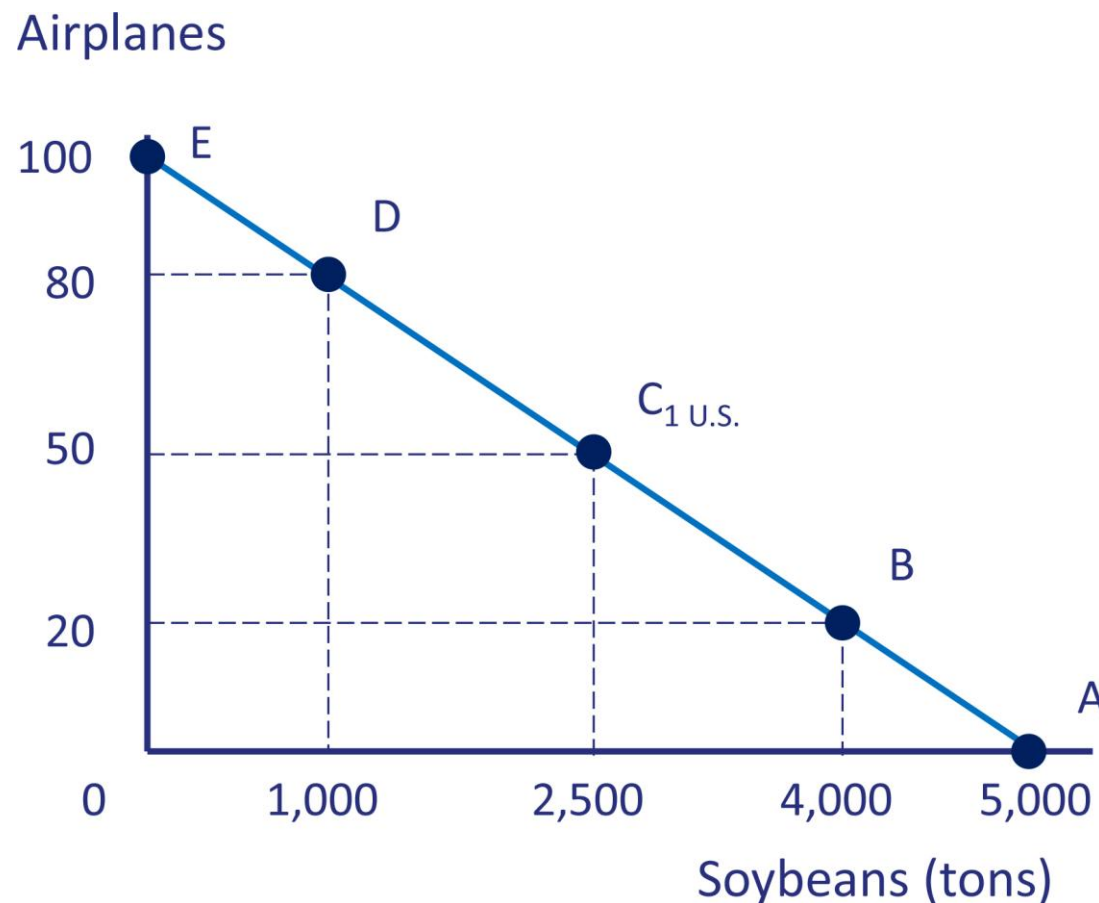
- The U.S. PPF
 - The U.S. economy has 50,000 labor hours per month available for production
 - Produces only two goods—airplanes and soybeans
 - To produce 1 airplane requires 500 labor hours
 - To produce 1 ton of soybeans requires 10 labor hours

Example 1: The U.S. PPF (2 of 3)

	Employment of Labor Hours	Employment of Labor Hours	Production	Production
	Airplanes	Soybeans	Airplanes	Soybeans
A	50,000	0	100	0
B	40,000	10,000	80	1,000
C	25,000	25,000	50	2,500
D	10,000	40,000	20	4,000
E	0	50,000	0	5,000

Example 1: The U.S. PPF (3 of 3)

- The United States has enough labor to produce any combination along the PPF
- Suppose the United States uses half its labor to produce each of the two goods
 - The U.S. production and consumption would be 50 airplanes and 2,500 tons of soybeans

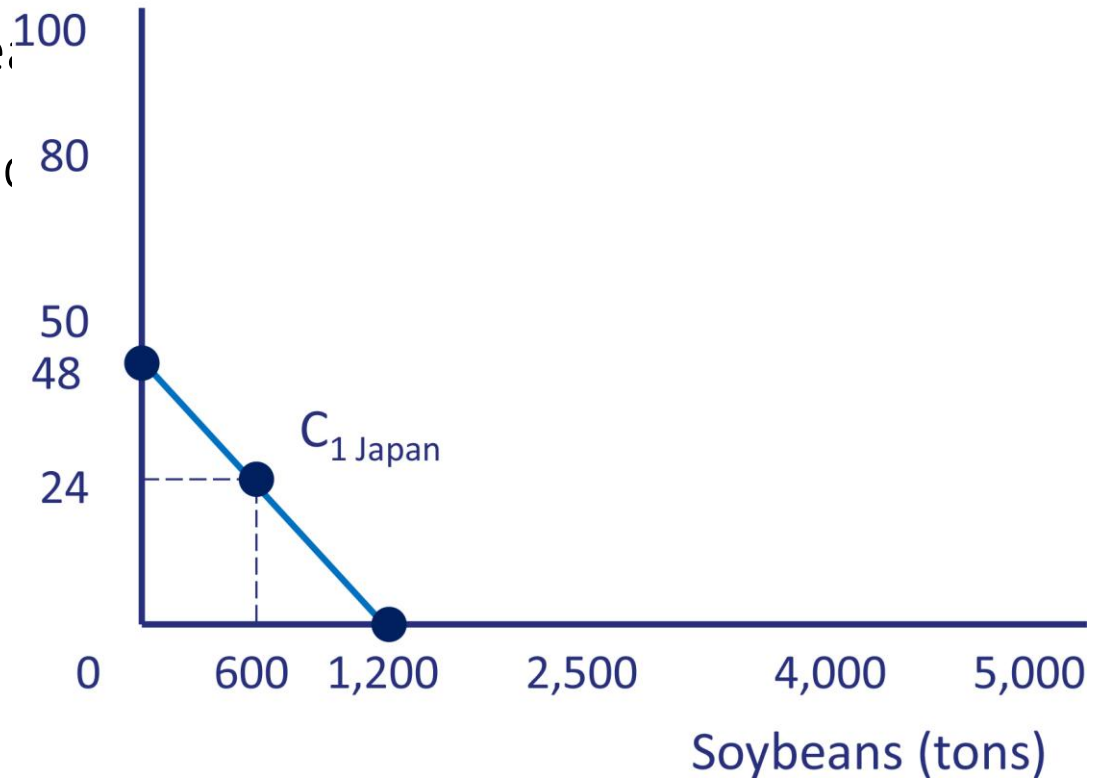


Active Learning 1: Japan's PPF

- Use the following information to draw Japan's PPF
 - Japan has 30,000 labor hours per month available for production
 - Produces only two goods—airplanes and soybeans
 - To produce 1 airplane requires 625 labor hours
 - To produce 1 ton of soybeans requires 25 labor hours
- Your graph should measure soybeans (tons) on the horizontal axis

Active Learning 1: Answers

- Japan has enough labor to produce any combination of **Airplanes** and **soybeans**
- Suppose Japan uses half its labor to produce each good
 - Japan's production and consumption would be at point C₁



Specialization and Trade

- Farmer Frank specializes in growing potatoes
 - More time growing potatoes
 - Less time raising cattle
- Rancher Ruby specializes in raising cattle
 - More time raising cattle
 - Less time growing potatoes
- Trade: 5 oz of meat for 15 oz of potatoes

Figure 2 How Trade Expands the Set of Consumption Opportunities (1 of 2)

The proposed trade offers Frank and Ruby a combination of meat and potatoes that would be impossible without trade. In panel (a), Frank consumes at point A* rather than point A. In panel (b), Ruby consumes at point B* rather than point B. Trade allows each to consume more meat and more potatoes.

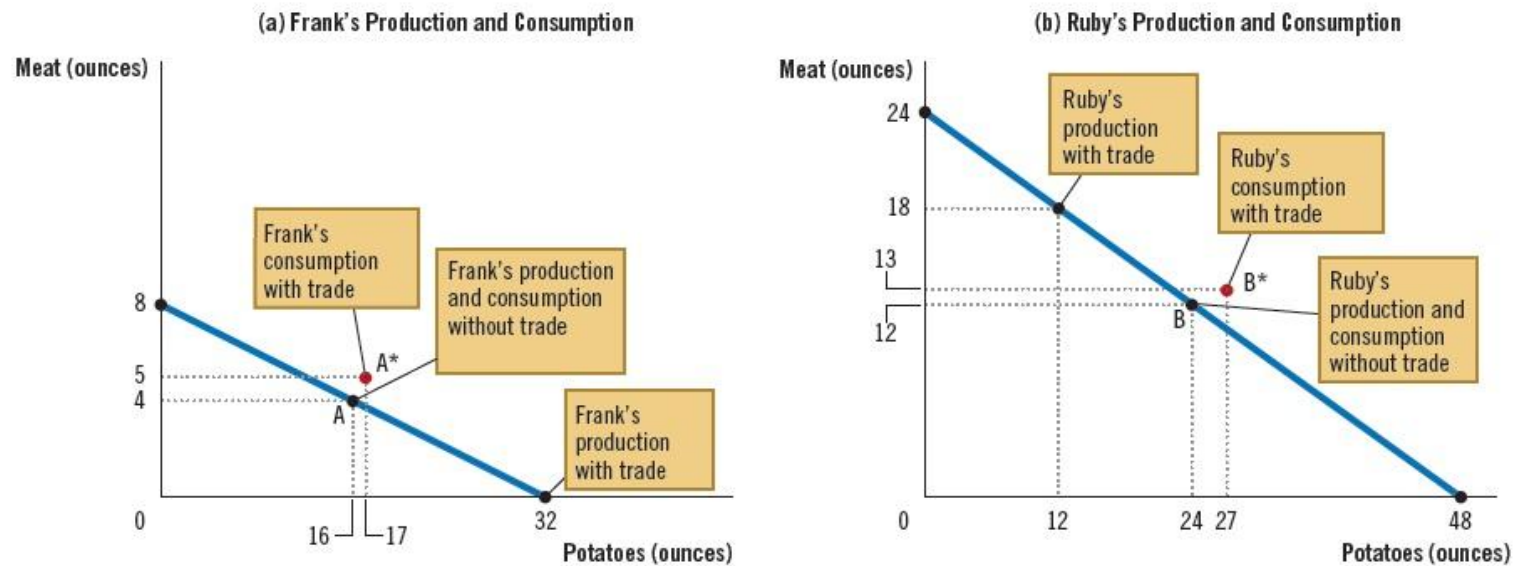


Figure 2 How Trade Expands the Set of Consumption Opportunities (2 of 2)

Trade allows each to consume more meat and more potatoes.

(c) The Gains from Trade: A Summary				
	Frank		Ruby	
	Meat	Potatoes	Meat	Potatoes
Without Trade:				
Production and Consumption	4 oz	16 oz	12 oz	24 oz
With Trade:				
Production	0 oz	32 oz	18 oz	12 oz
Trade	Gets 5 oz	Gives 15 oz	Gives 5 oz	Gets 15 oz
Consumption	5 oz	17 oz	13 oz	27 oz
GAINS FROM TRADE:				
Increase in Consumption	+1 oz	+1 oz	+1 oz	+3 oz

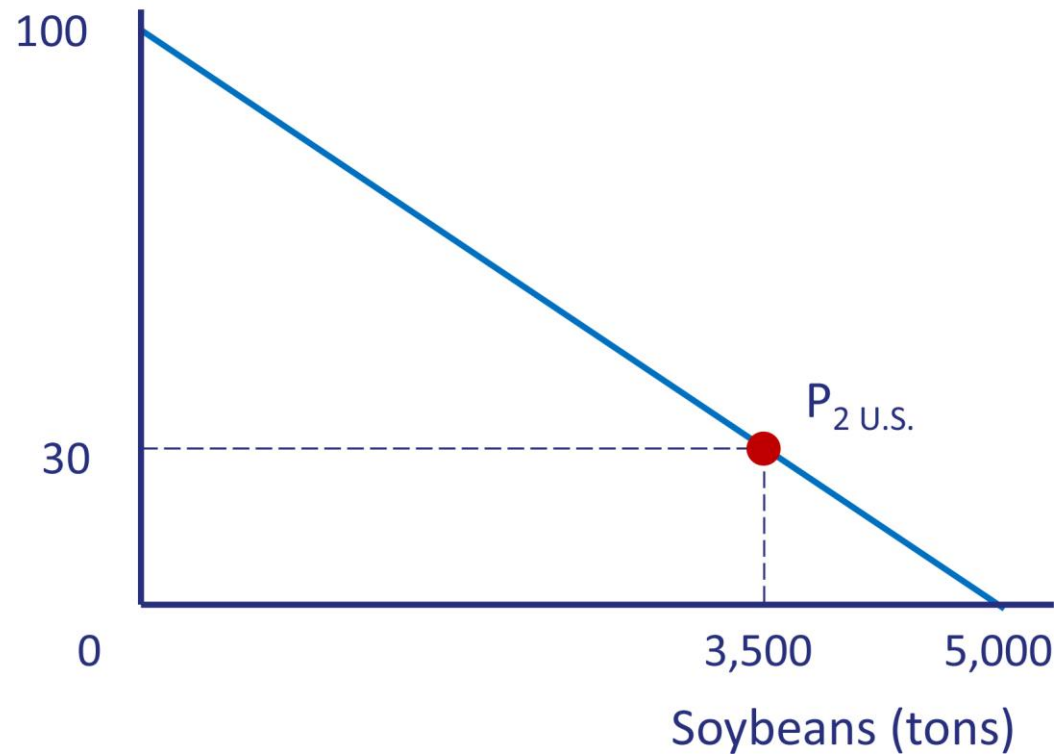
Active Learning 2: Production Under Trade

We continue Example 1 and Active Learning 1, but this time the two countries will choose different production points.

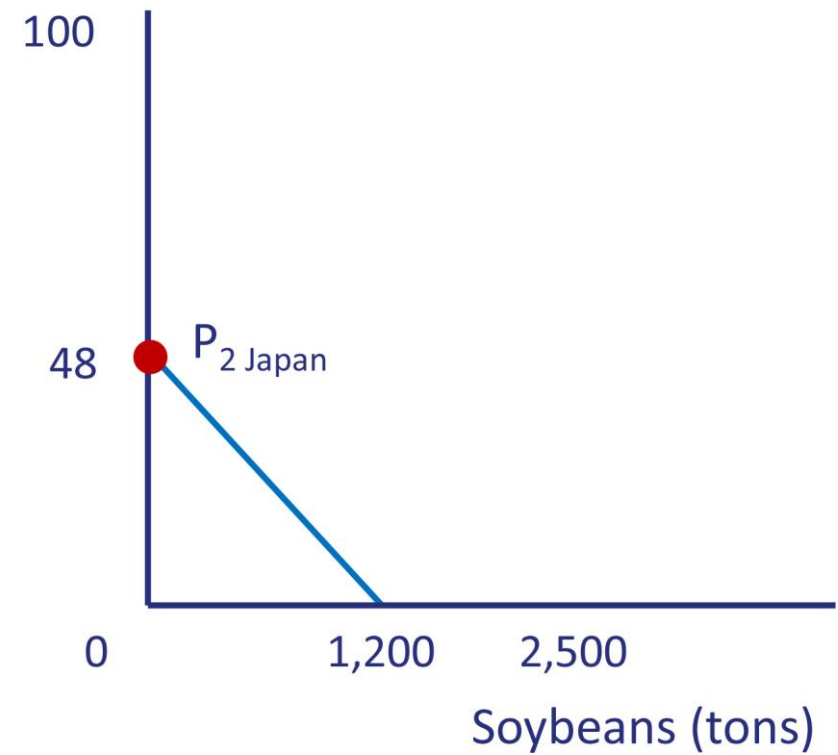
- A. The United States produces 3,500 tons of soybeans. How many airplanes can the United States produce with the remaining resources? Draw this point on the U.S. PPF.
- B. Japan produces 48 airplanes. How many tons of soybeans can Japan produce with the remaining resources? Draw this point on Japan's PPF.

Active Learning 2: Answers

Airplanes



Airplanes



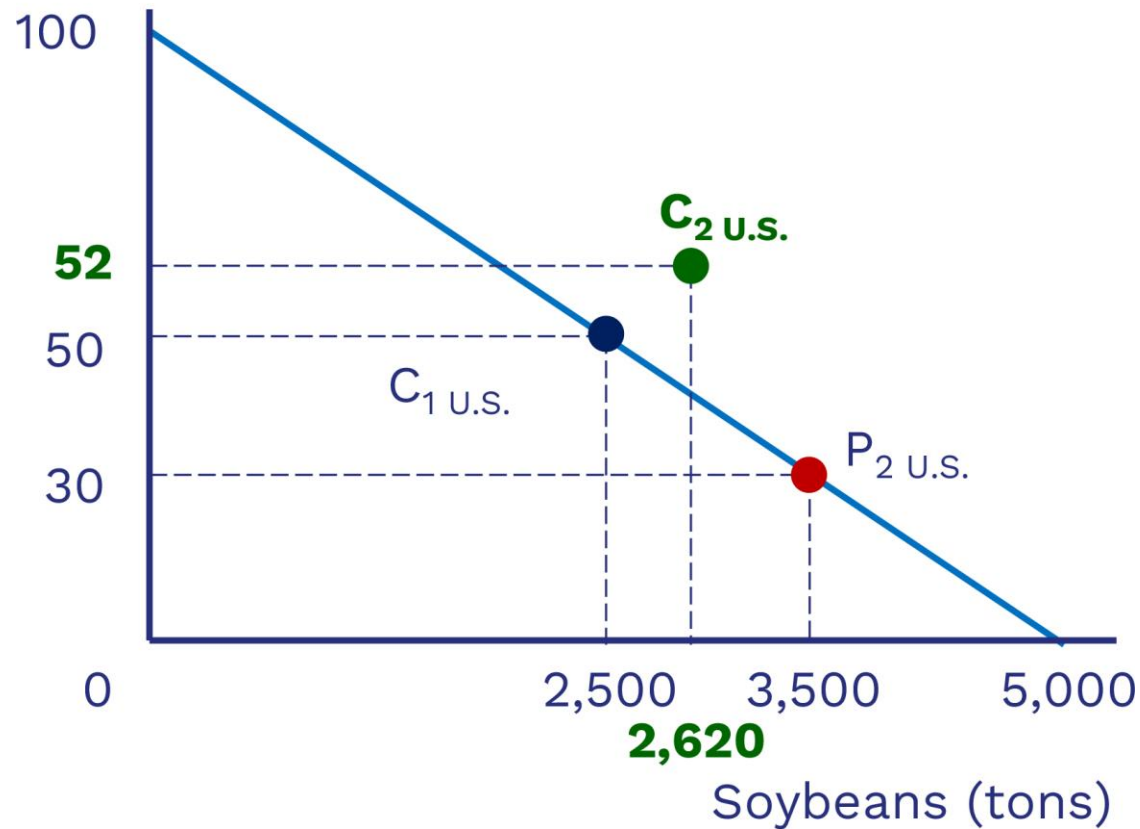
Active Learning 3: Consumption Under Trade

We continue Active Learning 2, but this time the two countries will be able to trade: 22 airplanes for 880 tons of soybeans

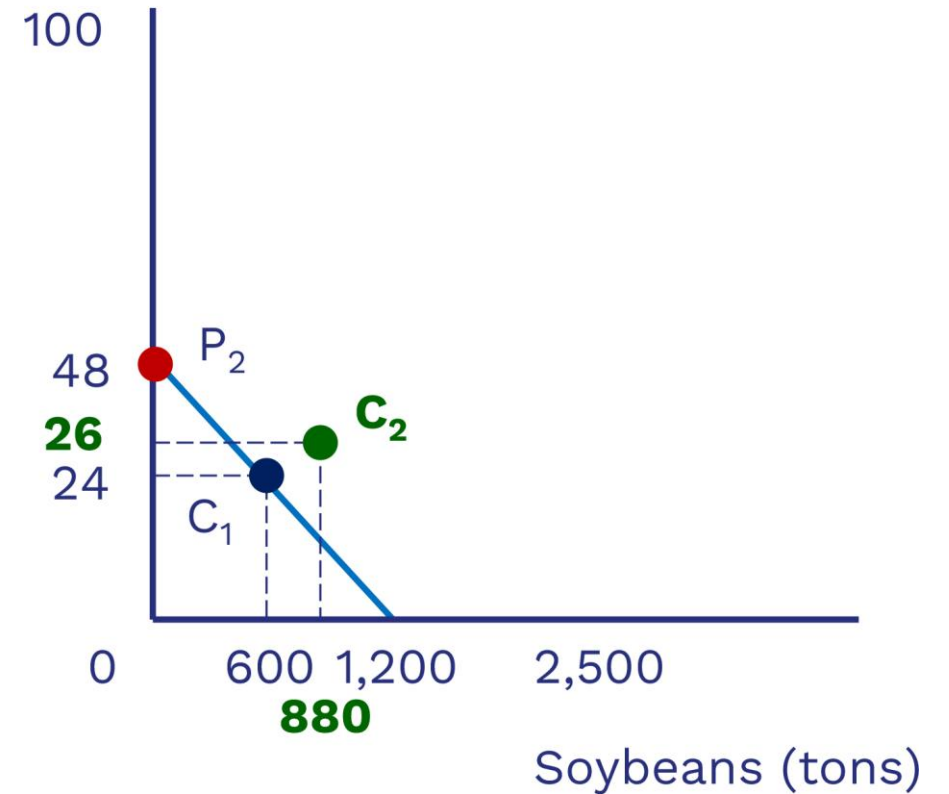
- A. The United States exports 880 tons of soybeans and imports 22 airplanes. How much of each good is consumed in the United States? Plot this combination on the U.S. PPF.
- B. Japan exports 22 airplanes and imports 880 tons of soybeans. How much of each good is consumed in Japan? Plot this combination on Japan's PPF.

Active Learning 3: Answers

Airplanes



Airplanes



3.2 Comparative Advantage: The Driving Force of Specialization

Absolute Advantage

- **Absolute advantage***
 - The ability to produce a good using fewer inputs than another producer
 - To produce 1 oz of meat—Ruby needs 20 minutes, Frank needs 60 minutes
 - To produce 1 oz of potatoes—Ruby needs 10 minutes, Frank needs 15 minutes
- Ruby has an absolute advantage in producing both meat and potatoes because she requires less time than Frank to produce a unit of either good

*Words accompanied by an asterisk are key terms from the chapter.

Opportunity Cost and Comparative Advantage

- **Opportunity cost***
 - Whatever must be given up to obtain some item
- Ruby—10 min. to grow 1 ounce of potatoes, 20 min. to produce 1 ounce of meat
 - Opportunity cost of producing 1 ounce of potatoes is $\frac{1}{2}$ ounce of meat
- Frank—15 min. to grow 1 oz potatoes, 60 min. to produce 1 oz meat
 - Opportunity cost of producing 1 ounce of potatoes is $\frac{1}{4}$ ounce of meat

*Words accompanied by an asterisk are key terms from the chapter.

Table 1 The Opportunity Cost of Meat and Potatoes

	Opportunity Cost	Opportunity Cost
	1 oz of Meat	1 oz of Potatoes
Frank the farmer	4 oz potatoes	$\frac{1}{4}$ oz meat
Ruby the rancher	2 oz potatoes	$\frac{1}{2}$ oz meat

Comparative Advantage (1 of 2)

- **Comparative advantage***
 - The ability to produce a good at a lower opportunity cost than another producer
- One person
 - Can have an absolute advantage in both goods
 - Cannot have a comparative advantage in both goods

*Words accompanied by an asterisk are key terms from the chapter.

Comparative Advantage (2 of 2)

- Opportunity cost of one good
 - Inverse of the opportunity cost of the other good
 - If a person's opportunity cost of one good is relatively high, opportunity cost of the other good must be relatively low
- For different opportunity costs
 - One person has comparative advantage in one good
 - The other person has comparative advantage in the other good

Comparative Advantage and Trade

- Gains from specialization and trade
 - Based on comparative advantage
 - Total production in economy rises
- Gains from trade
 - Reflected in the implicit prices that the trading partners pay each other
- Trade can benefit everyone because it allows people to specialize in the activities in which they have a comparative advantage

The Price of the Trade

- For both parties to gain from trade, the price at which they trade must lie between their opportunity costs

3.3 Applications of Comparative Advantage

Should Naomi Osaka Mow Her Own Lawn?

- In 2 hours, Osaka can mow her lawn, or
 - Film a TV commercial, earn \$30,000
- In 4 hours, Hari can mow Osaka's lawn, or
 - Work at McDonald's, earn \$50
- Gains from trade
 - As long as Osaka pays Hari more than \$50 and less than \$30,000, both are better off

Should the United States Trade with Other Countries? (1 of 4)

- **Imports***

- Goods produced abroad and sold domestically

- **Exports***

- Goods produced domestically and sold abroad

*Words accompanied by an asterisk are key terms from the chapter.

Should the United States Trade with Other Countries? (2 of 4)

- United States and Japan
 - Each produces food and cars
 - One American worker, in one month, can produce
 - One car or 2 tons of food
 - One Japanese worker, in one month, can produce
 - One car or 1 ton of food

Should the United States Trade with Other Countries? (3 of 4)

- Opportunity cost of a car
 - 2 tons of food in the United States, 1 ton of food in Japan
 - Japan has a comparative advantage in producing cars
- Opportunity cost of a ton of food
 - 1 car in Japan, $\frac{1}{2}$ car in the United States
 - The United States has a comparative advantage in producing food

Should the United States Trade with Other Countries? (4 of 4)

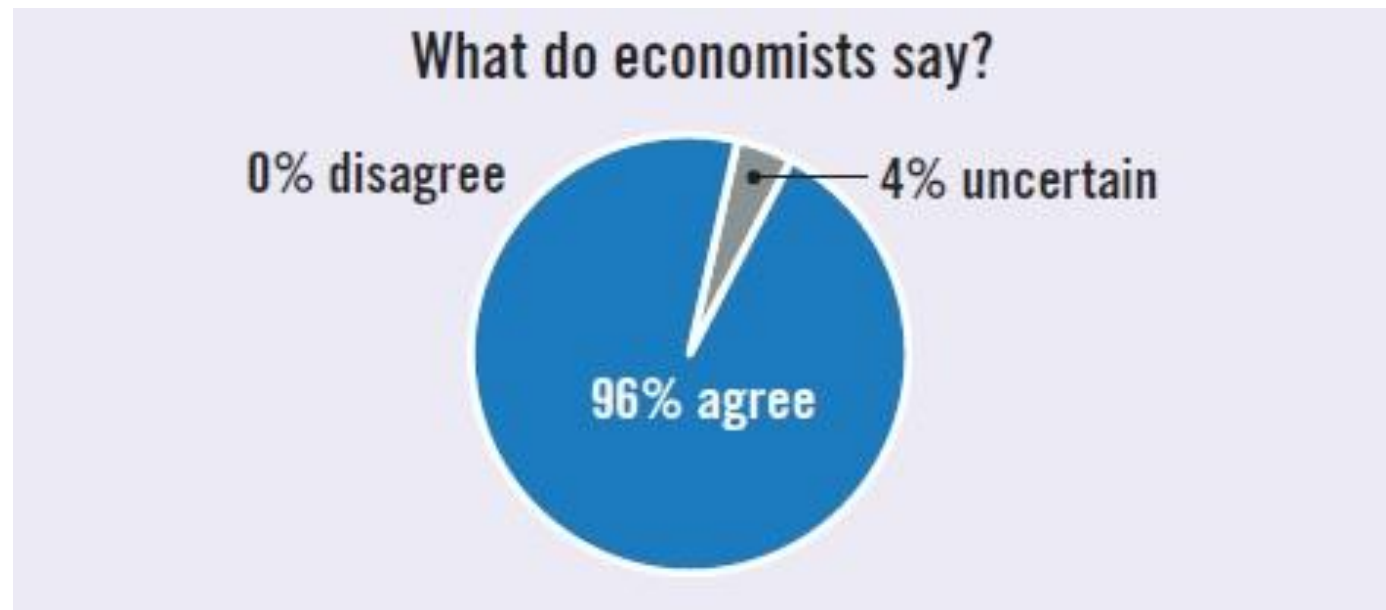
- Principle of comparative advantage
 - Each good should be produced by the country with the smaller opportunity cost of producing that good
 - Japan should produce more cars than it wants for its own use and export some of them to the United States
 - The United States should produce more food than it wants to consume and export some to Japan
- Through specialization and trade, both countries have more food and more cars

3.4 Conclusion

- The principle of comparative advantage shows that trade can make everyone better off
- How is it possible?
 - How do free societies coordinate the diverse activities of all the people involved in their economies?
 - What ensures that goods and services will get from those who should be producing them to those who should be consuming them?
- Most economies allocate resources using the market forces of supply and demand

Ask the Experts: Trade between China and the United States - B

“Some Americans who work in the production of competing goods, such as clothing and furniture, are made worse off by trade with China.”



Source: IGM Economic Experts Panel, June 19, 2012.

Think-Pair-Share Activity

- You are watching an election debate on television. A candidate says, “We need to stop the flow of foreign steel into our country. If we place a tariff on imports of steel, our domestic steel production will rise and the United States will be better off.”
 - A. Will the United States be better off if we limit steel imports? Explain.
 - B. Will anyone in the United States be better off if we limit steel imports? Explain.
 - C. In the real world, does every person in the country gain when restrictions on imports are reduced? Explain.

Self-Assessment

- What exactly do people gain when they trade with one another?
- Compared to a roommate, are there activities in which you have an absolute advantage? A comparative advantage?

Chapter 4: The Market Forces of Supply and Demand

Chapter Objectives (1 of 3)

By the end of this chapter, you should be able to:

- Given a scenario, determine if a market is competitive.
- Construct a demand curve using a given demand schedule.
- Describe the relationship between price and quantity demanded using the law of demand.
- Determine if a given scenario will cause a movement along or a shift of a good's demand curve.
- Explain how a change in a demand determinant impacts a good's demand curve.

Chapter Objectives (2 of 3)

- Determine market demand using information about individuals' demand.
- Determine if a given scenario will cause a movement along or a shift of a good's supply curve.
- Explain how the price of a good impacts the demand for its complements and substitutes.
- Construct a supply curve using a given supply schedule.
- Describe the relationship between price and quantity supplied using the law of supply.
- Explain how a change in a supply determinant impacts a good's supply curve.

Chapter Objectives (3 of 3)

- Determine market supply using information about individual firms' supply.
- Determine the equilibrium price and quantity using the supply and demand model.
- Explain how simultaneous changes in demand and supply impact market equilibrium.
- Explain how changes in demand impact market equilibrium.
- Explain how changes in supply impact market equilibrium.
- Explain how price changes eliminate a surplus or shortage.

4.1 Markets and Competition

Supply and Demand

- Supply and demand
 - Words economists use most often
 - Forces that make market economies
 - Refer to the behavior of people as they interact in competitive markets

What Is a Market?

- **Market***
 - A group of buyers and sellers of a particular good or service
- Buyers as a group
 - Determine the demand for the product
- Sellers as a group
 - Determine the supply of the product

*Words accompanied by an asterisk are key terms from the chapter.

What Is Competition?

- **Competitive market***
 - Market in which there are many buyers and many sellers
 - Each has a negligible impact on market price
- Price and quantity sold are determined by all buyers and sellers as they interact in the marketplace

*Words accompanied by an asterisk are key terms from the chapter.

Perfectly Competitive Market

- Goods offered for sale are all exactly the same
- Buyers and sellers are numerous
 - No single buyer or seller has any influence over the market price
 - Must accept the price the market determines (price takers)
- At the market price
 - Buyers can buy all they want
 - Sellers can sell all they want

Other Markets

- Monopoly
 - The only seller in the market
 - Sets the price
- Other Markets
 - Between perfect competition and monopoly

4.2 Demand

Law of Demand

- **Law of demand***
 - Other things being equal, when the price of a good rises, the quantity demanded falls, and when the price falls, the quantity demanded rises
- **Quantity demanded***
 - Amount of a good that buyers are willing and able to purchase

*Words accompanied by an asterisk are key terms from the chapter.

Individual Demand

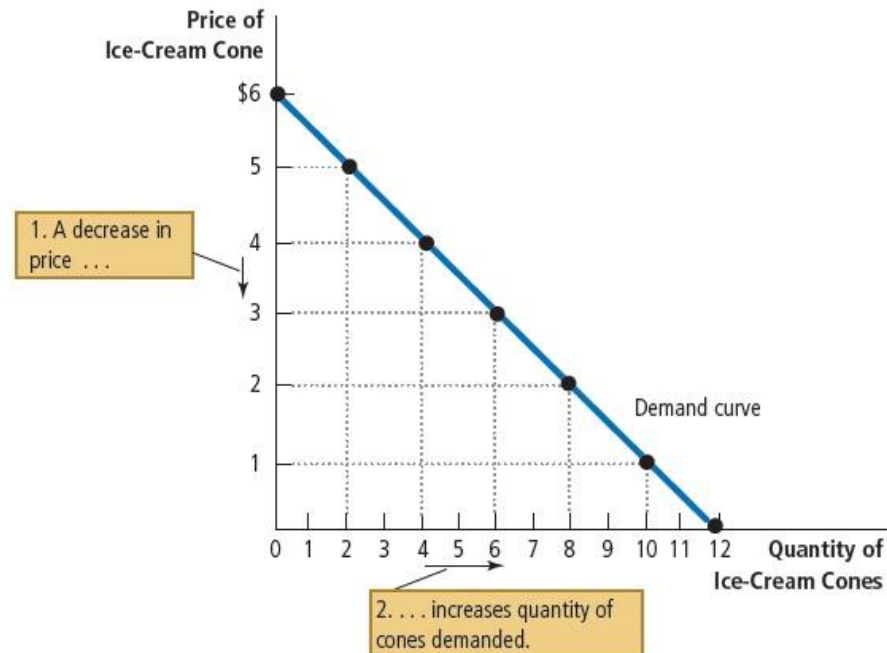
- Individual demand
 - An individual's demand for a product
- **Demand schedule***
 - A table that shows the relationship between the price of a good and the quantity demanded
- **Demand curve***
 - A graph of the relationship between the price of a good and the quantity demanded

*Words accompanied by an asterisk are key terms from the chapter.

Figure 1 Catherine's Demand Schedule and Demand Curve

The demand schedule is a table that shows the quantity demanded at each price. The demand curve, which graphs this schedule, illustrates how the quantity demanded changes as the price varies. Because a lower price increases the quantity demanded, the demand curve slopes downward.

Price of Ice-Cream Cone	Quantity of Cones Demanded
\$0	12 cones
1	10
2	8
3	6
4	4
5	2
6	0



Market Demand

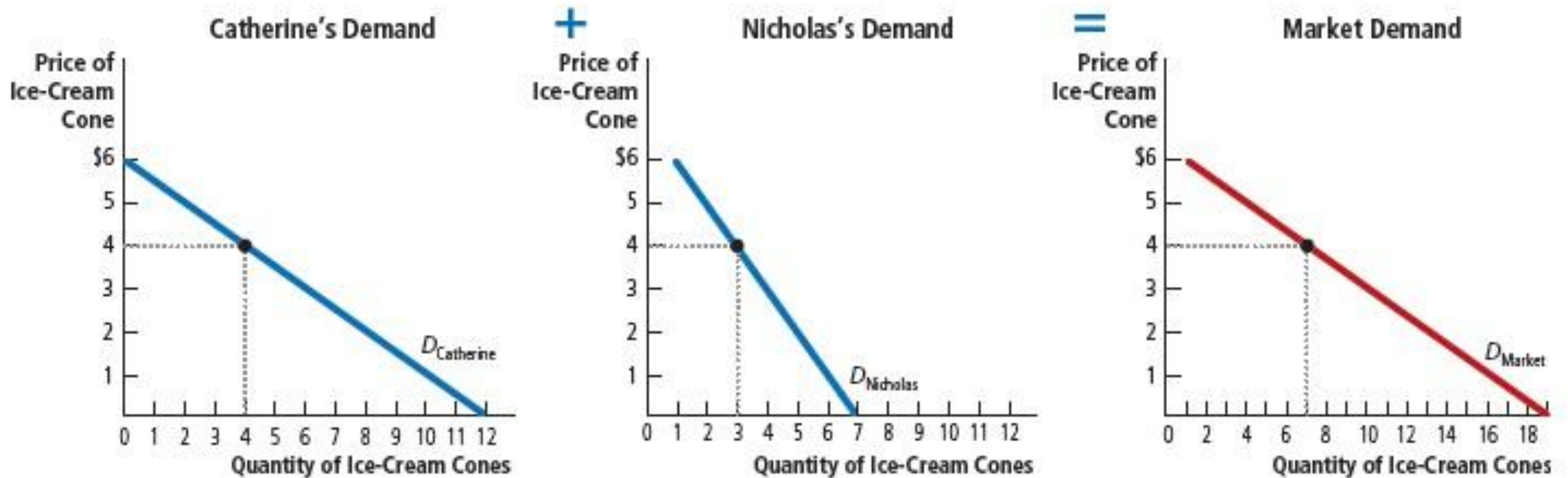
- Market demand
 - The sum of all the individual demands for a particular good or service
- Market demand curve
 - Shows how the total quantity demanded of a good varies as its price changes, holding constant all the other factors that affect consumer purchases

Figure 2 Market Demand as the Sum of Individual Demands (1 of 2)

The quantity demanded in a market is the sum of the quantities demanded by all the buyers at each price. Thus, the market demand curve is found by adding the individual demand curves horizontally. At a price of \$4, Catherine demands 4 ice-cream cones, and Nicholas demands 3, so the quantity demanded in the market at this price is 7 cones.

Price of Ice-Cream Cone	Catherine		Nicholas		Market
\$0	12	+	7	=	19 cones
1	10		6		16
2	8		5		13
3	6		4		10
4	4		3		7
5	2		2		4
6	0		1		1

Figure 2 Market Demand as the Sum of Individual Demands (2 of 2)

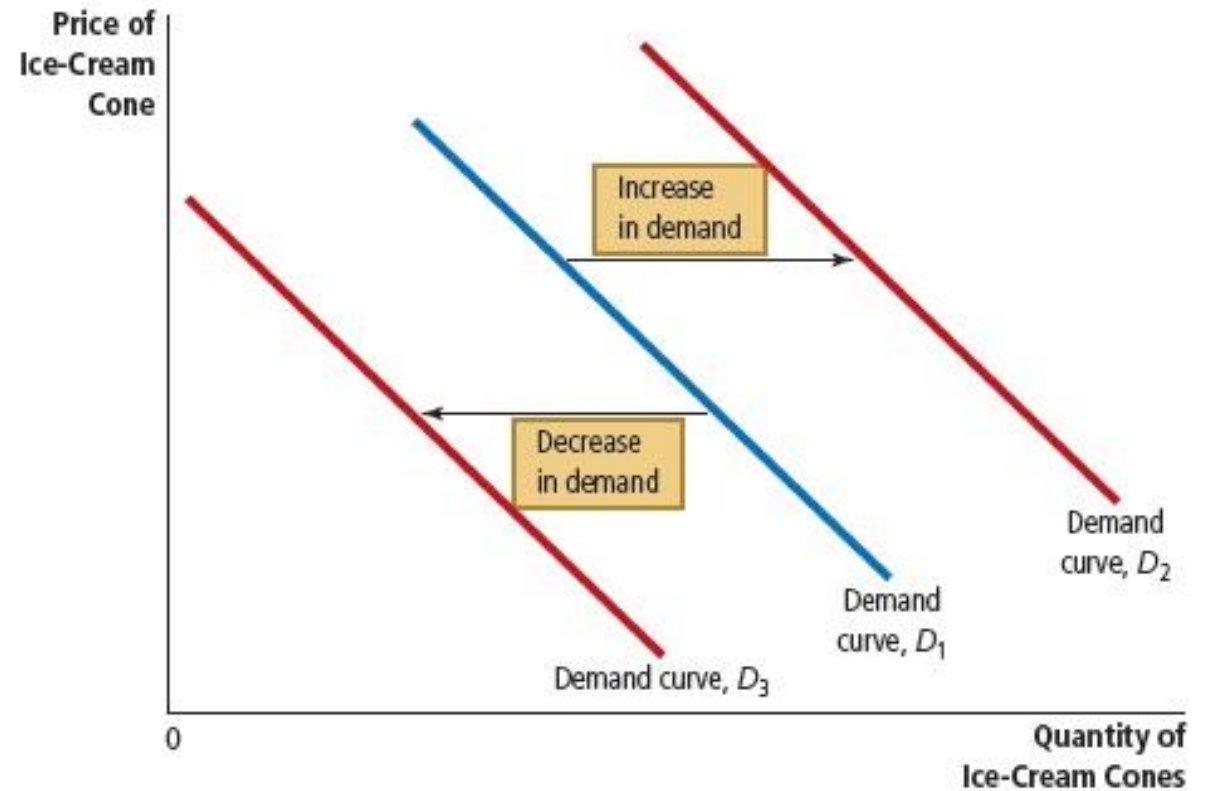


Shifts in the Demand Curve

- Market demand curve
 - Holds other things constant
 - Does not need to be stable over time
- Demand curve shifts
 - Change that increases quantity buyers want to purchase at any price shifts demand curve to the right
 - Change that decreases quantity buyers want to purchase at any price shifts demand curve to the left

Figure 3 Shifts in the Demand Curve

- A change that increases the quantity that buyers want to purchase at any price shifts the demand curve to the right.
- A change that decreases the quantity that buyers want to purchase at any price shifts the demand curve to the left.



Variables That Influence Buyers (1 of 3)

- Income
 - **Normal good***
 - An increase in income leads to an increase in demand
 - **Inferior good***
 - An increase in income leads to a decrease in demand

*Words accompanied by an asterisk are key terms from the chapter.

Variables That Influence Buyers (2 of 3)

- Prices of related goods
 - **Substitutes***
 - Pairs of goods that are used in place of each other
 - Increase in the price of one leads to an increase in the demand for the other
 - **Complements***
 - Pairs of goods that are used together
 - Increase in the price of one leads to a decrease in the demand for the other

*Words accompanied by an asterisk are key terms from the chapter.

Variables That Influence Buyers (3 of 3)

- Tastes
 - Unique, affected by historical and psychological forces
- Expectations
 - Future changes in income
 - Future changes in prices
- Number of buyers
 - Market demand depends on how many buyers there are in the market

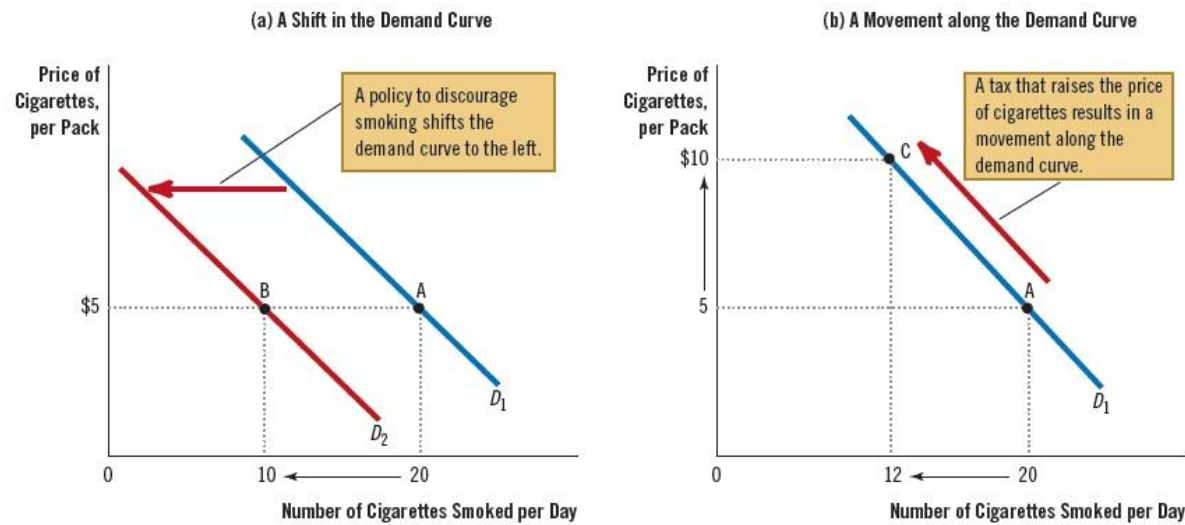
Table 1 Variables That Influence Buyers

- This table lists the variables that affect how much of any good consumers choose to buy.
- Notice the special role that the price of the good plays: A change in that price represents a movement along the demand curve, while a change in one of the other variables shifts the curve.

Variable	A Change in This Variable . . .
Price of the good itself	Represents a movement along the demand curve
Income	Shifts the demand curve
Prices of related goods	Shifts the demand curve
Tastes	Shifts the demand curve
Expectations	Shifts the demand curve
Number of buyers	Shifts the demand curve

Figure 4 Shifts in the Demand Curve versus Movements along the Demand Curve

When warnings on cigarette packages persuade smokers to smoke less, the demand curve for cigarettes shifts to the left. In panel (a), the curve shifts from D_1 to D_2 . At a price of \$5 per pack, the quantity demanded falls from 20 to 10 cigarettes per day, as reflected by the shift from point A to point B. By contrast, when a tax raises the price of cigarettes, the demand curve does not shift. Instead, there is a movement to a different point on the demand curve. In panel (b), when the price rises from \$5 to \$10, the quantity demanded falls from 20 to 12 cigarettes per day, as reflected by the movement from point A to point C.

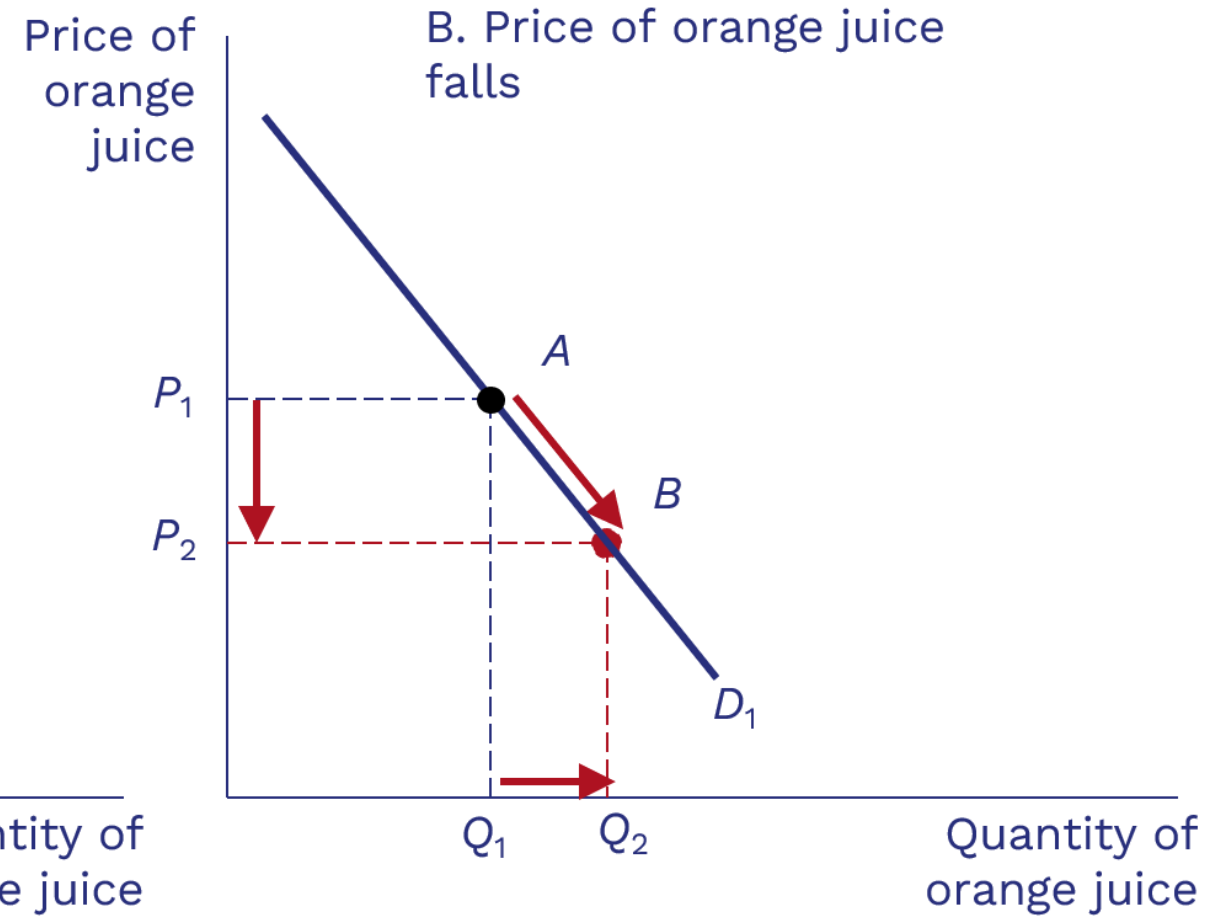
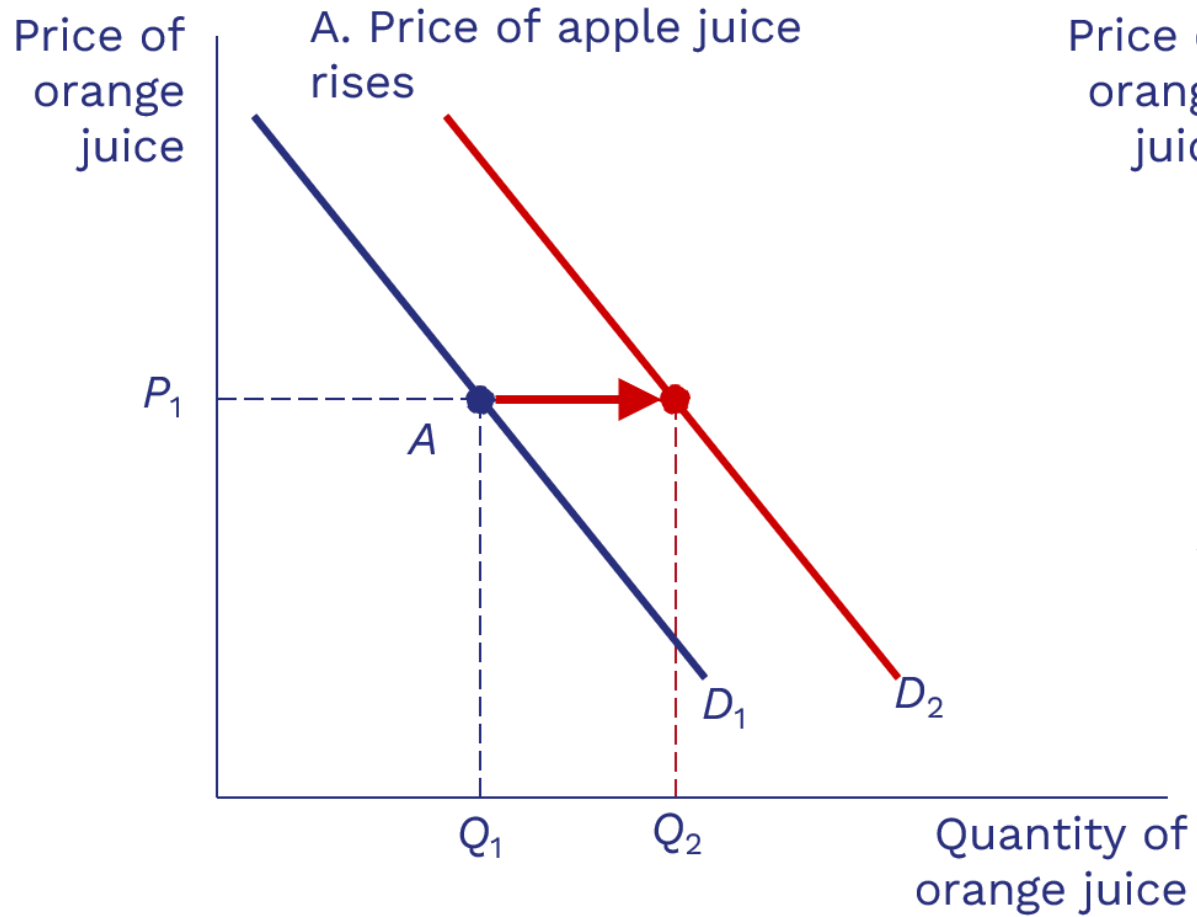


Active Learning 1: The Demand Curve, D

Draw the demand curve for orange juice, D_1 , and choose a point A (P_1 , Q_1) on the demand curve. What happens in these following scenarios? Why?

- A. Price of apple juice rises
- B. Price of orange juice falls

Active Learning 1: Answers



4.3 Supply

Law of Supply

- **Law of supply***
 - Other things being equal, when the price of a good rises, the quantity supplied also rises, and when the price falls, the quantity supplied falls as well
- **Quantity supplied***
 - Amount of a good that sellers are willing and able to sell

*Words accompanied by an asterisk are key terms from the chapter.

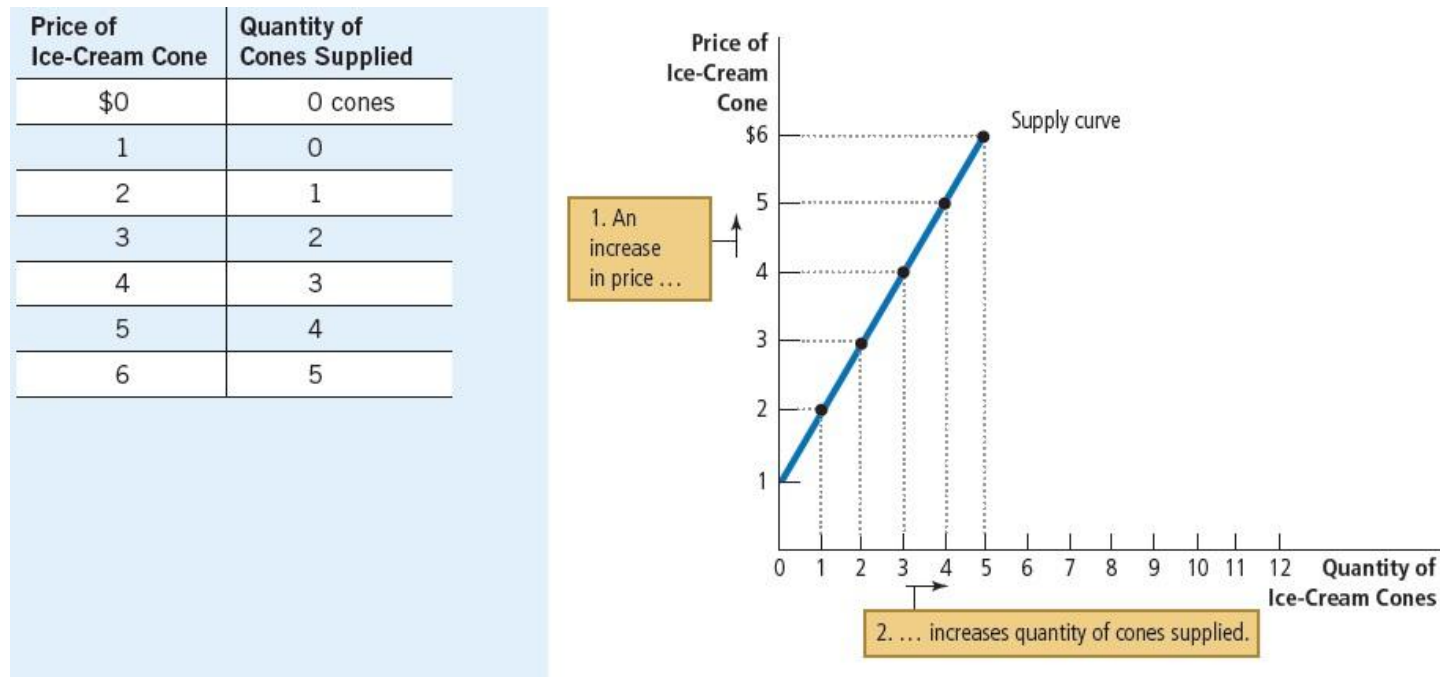
Individual Supply

- Individual supply
 - A seller's supply for a product
- **Supply schedule***
 - A table that shows the relationship between the price of a good and the quantity supplied
- **Supply curve***
 - A graph of the relationship between the price of a good and the quantity supplied

*Words accompanied by an asterisk are key terms from the chapter.

Figure 5 Ben's Supply Schedule and Supply Curve

The supply schedule is a table that shows the quantity supplied at each price. The supply curve, which graphs the supply schedule, illustrates how the quantity supplied changes as a good's price varies. Because a higher price increases the quantity supplied, the supply curve slopes upward.



Market Supply

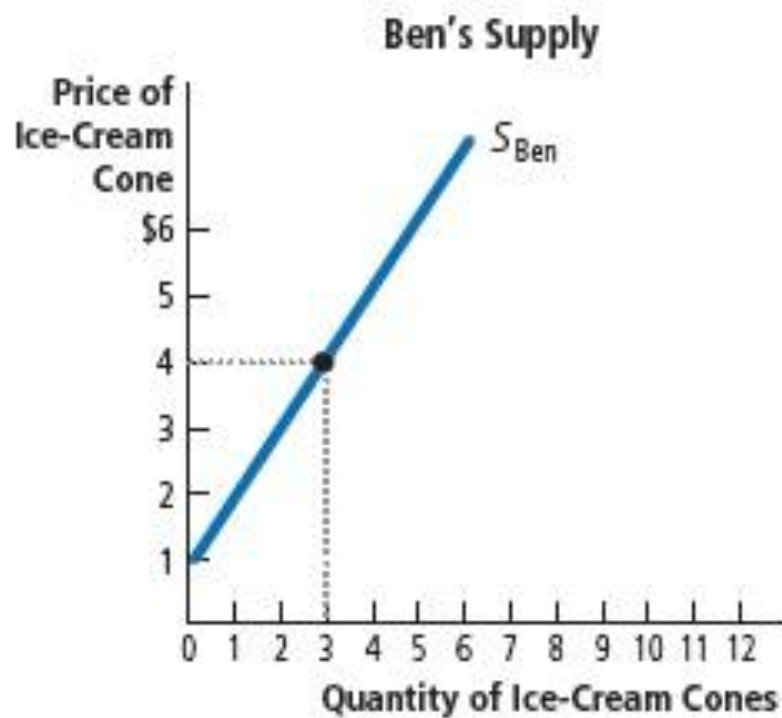
- Market supply
 - The sum of the supplies of all sellers
- Market supply curve
 - Shows how the total quantity supplied varies as the price varies, holding constant all other factors that influence producers' decisions about how much to sell

Figure 6 Market Supply as the Sum of Individual Supplies (1 of 2)

The quantity supplied in a market is the total quantity supplied by all sellers at each price. You can build the market supply curve by adding the individual supply curves horizontally. At a price of \$4, Ben supplies 3 ice-cream cones and Jerry supplies 4 ice-cream cones, so the total quantity supplied at the price of \$4 is 7 cones.

Price of Ice-Cream Cone	Ben		Jerry		Market
\$0	0	+	0	=	0 cones
1	0		0		0
2	1		0		1
3	2		2		4
4	3		4		7
5	4		6		10
6	5		8		13

Figure 6 Market Supply as the Sum of Individual Supplies (2 of 2)

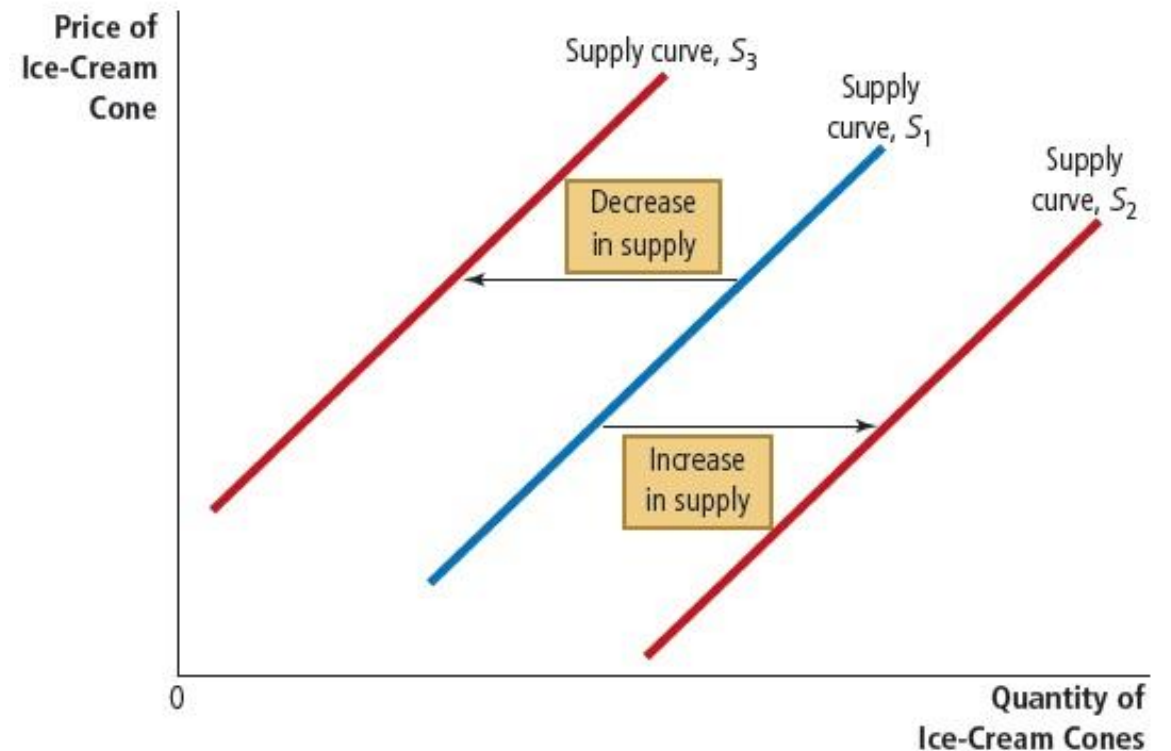


Shifts in the Supply Curve

- Market supply curve
 - Holds other things constant
 - Does not need to be stable over time
- Supply curve shifts
 - Change that increases quantity sellers want to sell at any price shifts supply curve to the right
 - Change that decreases quantity sellers want to sell at any price shifts supply curve to the left

Figure 7 Shifts in the Supply Curve

- A change that raises the quantity that sellers want to produce at any price shifts the supply curve to the right.
- A change that lowers the quantity that sellers want to produce at any price shifts the supply curve to the left.



Variables That Influence Sellers (1 of 2)

- Input prices
 - The supply of a good moves in the opposite direction of the prices of inputs
- Technology
 - Technology for turning inputs into output
 - Advances in technology increase the supply

Variables That Influence Sellers (2 of 2)

- Expectations
 - Future changes in prices
- Number of sellers
 - Market supply depends on how many sellers there are in the market

Table 2 Variables That Influence Sellers

- This table lists the variables that affect how much of any good producers choose to sell.
- Notice the special role that the price of the good plays: A change in that price represents a movement along the supply curve, while a change in one of the other variables shifts the curve.

Variable	A Change in This Variable . . .
Price of the good itself	Represents a movement along the supply curve
Input prices	Shifts the supply curve
Prices of related goods	Shifts the supply curve
Technology	Shifts the supply curve
Expectations	Shifts the supply curve
Number of sellers	Shifts the supply curve

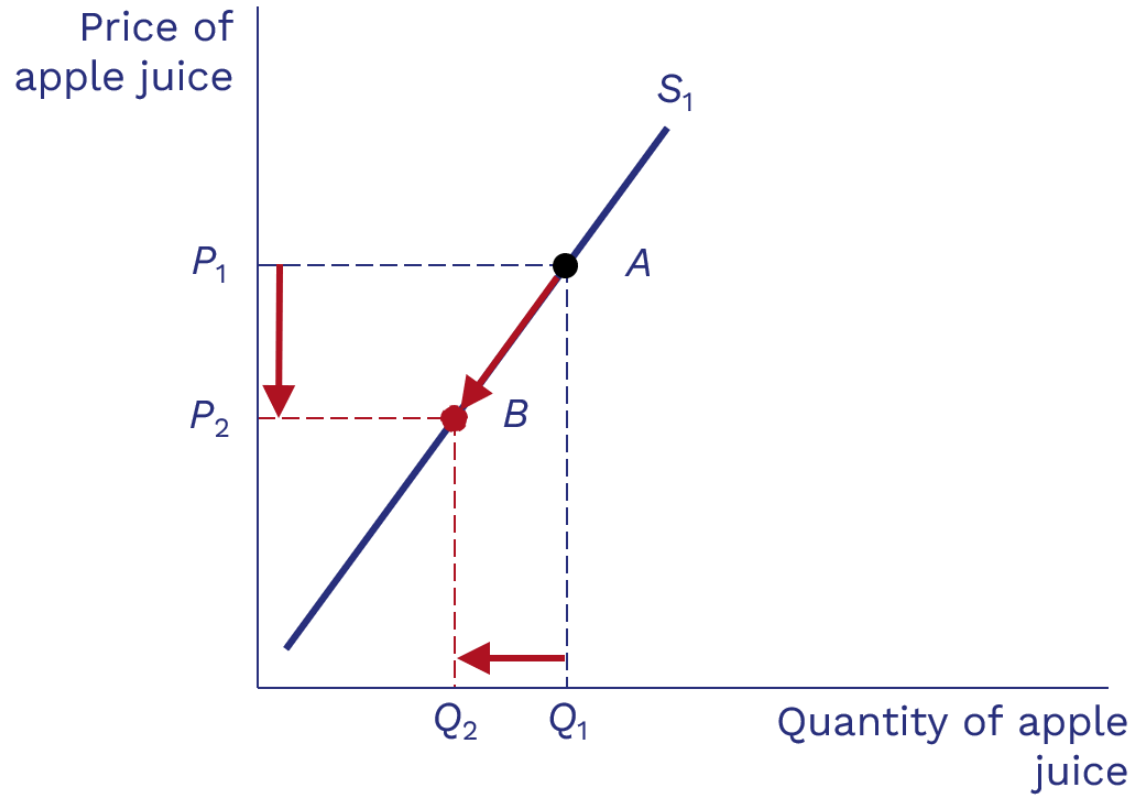
Active Learning 2: The Supply Curve, S

Draw a supply curve for apple juice, S_1 , and choose a point A (P_1, Q_1) on the supply curve. What happens to it in each of the following scenarios? Why?

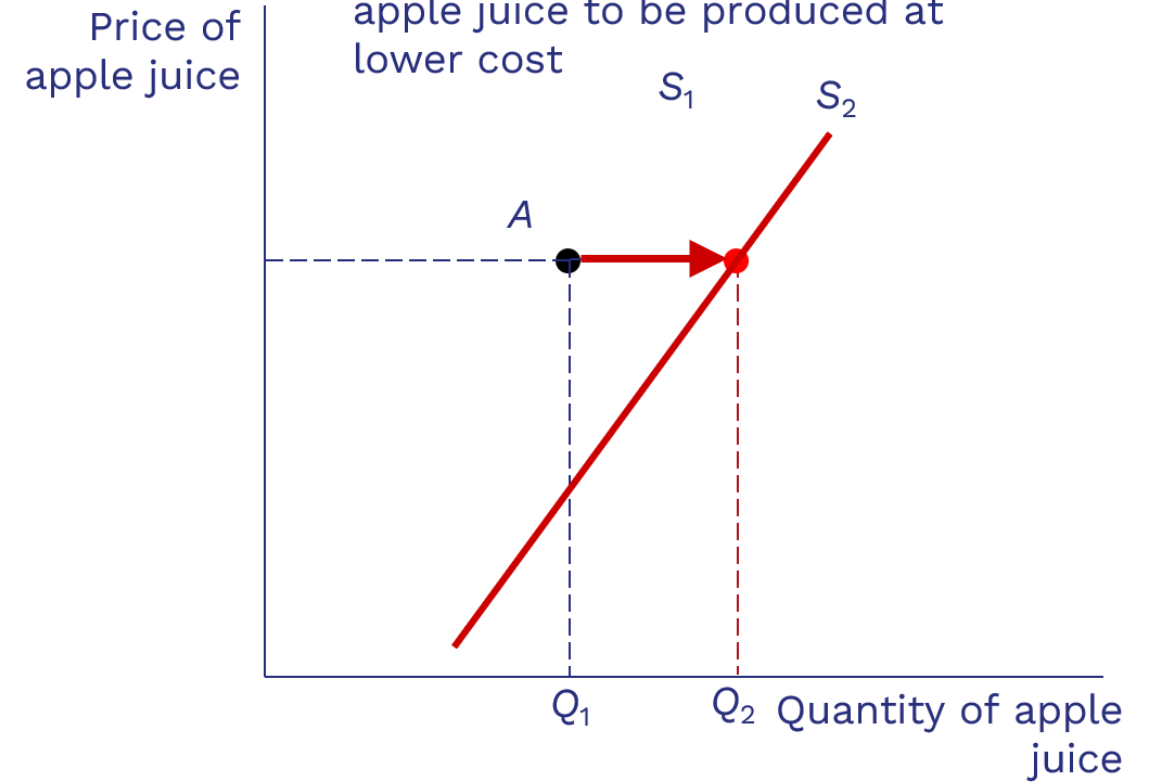
- A. Grocery stores cut the price of apple juice
- B. A technological advance allows apple juice to be produced at lower cost

Active Learning 2: Answers

A. Grocery stores cut the price of apple juice



B. A technological advance allows apple juice to be produced at lower cost



4.4 Supply and Demand Together

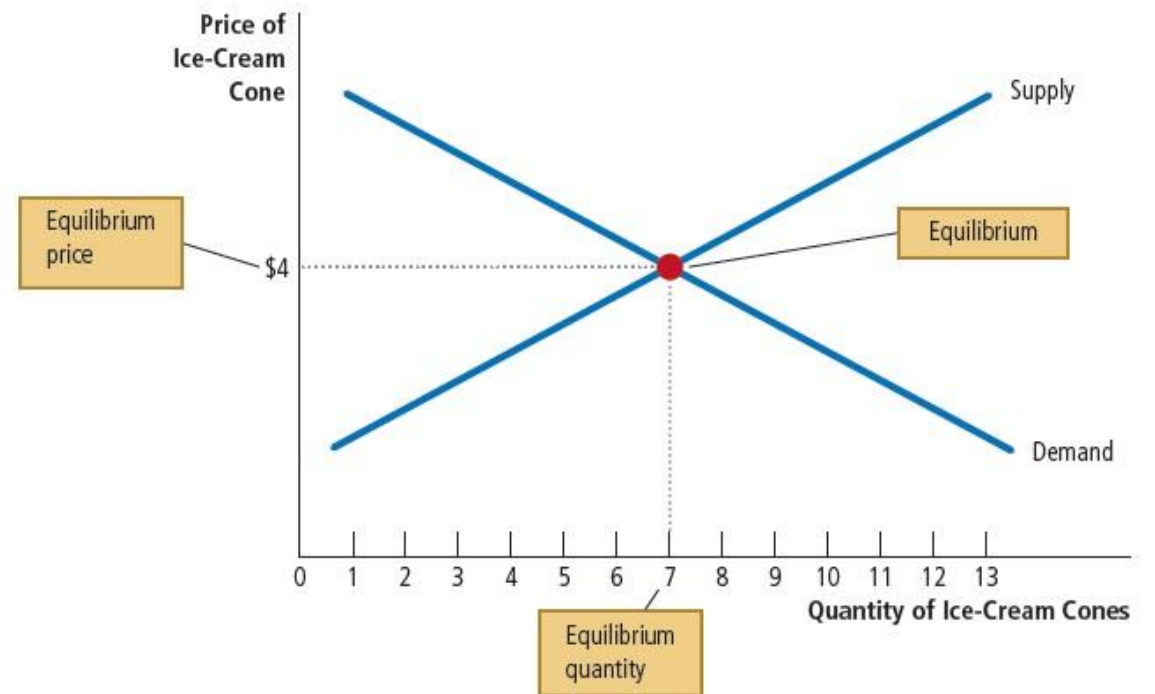
Equilibrium

- **Equilibrium***
 - Quantity of the good that buyers are willing and able to buy exactly balances quantity that sellers are willing and able to sell
- **Equilibrium price***
 - Balances the quantity supplied and quantity demanded
- **Equilibrium quantity***
 - Quantity supplied and quantity demanded at the equilibrium price

*Words accompanied by an asterisk are key terms from the chapter.

Figure 8 The Equilibrium of Supply and Demand

- The market's equilibrium is where the supply and demand curves intersect.
- At the equilibrium price, the quantity supplied equals the quantity demanded.
- Here, the equilibrium price is \$4.
- At this price, 7 ice-cream cones are supplied, and 7 are demanded.



Surplus

- **Surplus***
 - Quantity supplied is greater than quantity demanded
- Sellers respond by cutting prices
 - Increase quantity demanded and decrease quantity supplied
 - Changes represent movements along the supply and demand curves
 - Prices continue to fall until the market reaches the equilibrium

*Words accompanied by an asterisk are key terms from the chapter.

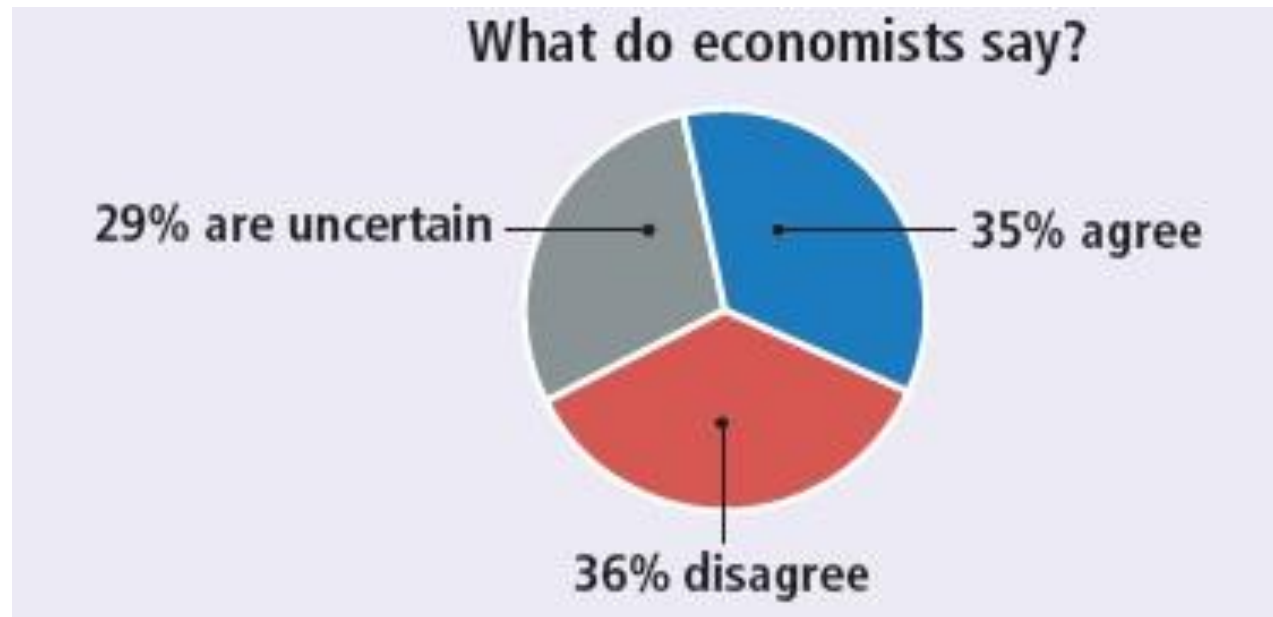
Shortage

- **Shortage***
 - Quantity demanded is greater than quantity supplied
- Sellers can raise prices without losing sales
 - Decrease quantity demanded and increase quantity supplied
 - Changes represent movements along the supply and demand curves
 - Prices continue to fall until the market reaches the equilibrium

*Words accompanied by an asterisk are key terms from the chapter.

Ask the Experts: Price Gouging

“Laws to prevent high prices for essential goods in short supply in a crisis would raise social welfare.”



Source: IGM Economic Experts Panel, May 26, 2020.

Law of Supply and Demand

- **Law of supply and demand***

- Claim that the price of any good adjusts to bring the quantity supplied and the quantity demanded of that good into balance
- In well-functioning markets, surpluses and shortages are only temporary because prices quickly move toward their equilibrium levels

*Words accompanied by an asterisk are key terms from the chapter.

Figure 9 Markets Not in Equilibrium

In panel (a), there is a surplus. Because the market price of \$5 is above the equilibrium price, the quantity supplied (10 cones) exceeds the quantity demanded (4 cones). Producers try to increase sales by cutting the price, moving it toward its equilibrium level. In panel (b), there is a shortage. Because the market price of \$3 is below the equilibrium price, the quantity demanded (10 cones) exceeds the quantity supplied (4 cones). With too many buyers chasing too few goods, producers raise the price. In both cases, the price adjustment moves the market toward the equilibrium of supply and demand.

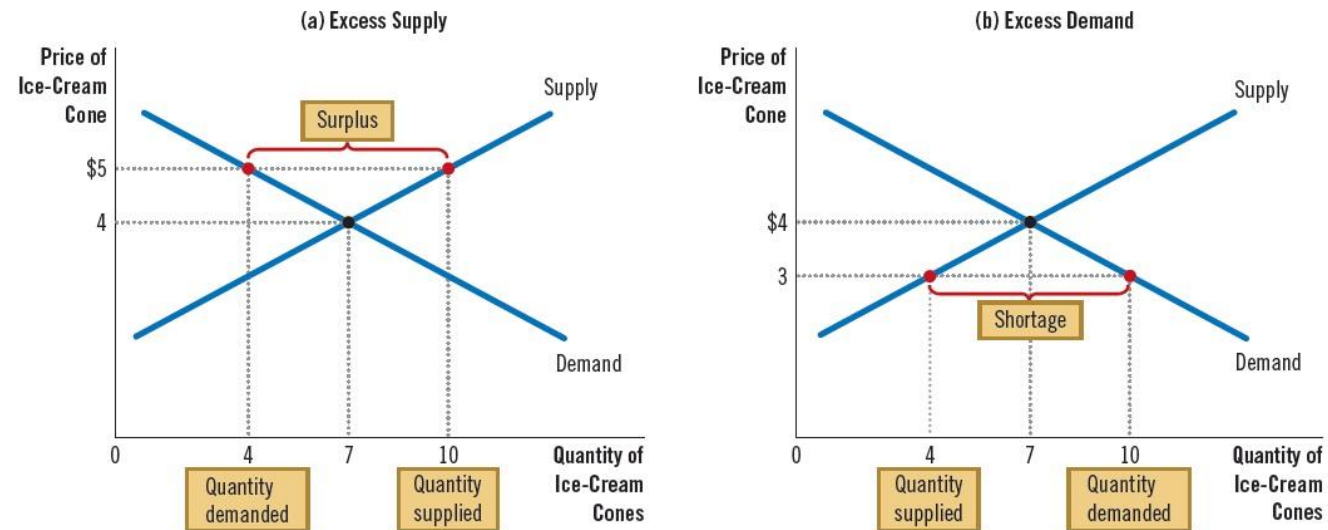


Table 3 Three Steps for Analyzing Changes in Equilibrium

1. Decide if the event shifts the supply or demand curve (or perhaps both).
2. Decide in which direction the curve shifts.
3. Use a supply-and-demand diagram to see how the shift changes the equilibrium price and quantity.

Shifts in Curves versus Movements along Them

Change in supply

- A shift in the supply curve

Change in demand

- A shift in the demand curve

Change in the quantity supplied

- A movement along a fixed supply curve

Change in the quantity demanded

- A movement along a fixed demand curve

Figure 10 How an Increase in Demand Affects the Equilibrium

- An event that raises the quantity demanded at any price shifts the demand curve to the right.
- The equilibrium price and quantity both rise.
- Here, an abnormally hot summer causes buyers to demand more ice cream.
- The demand curve shifts from D_1 to D_2 , causing the equilibrium price to increase from \$4 to \$5 and the equilibrium quantity to increase from 7 to 10 cones.

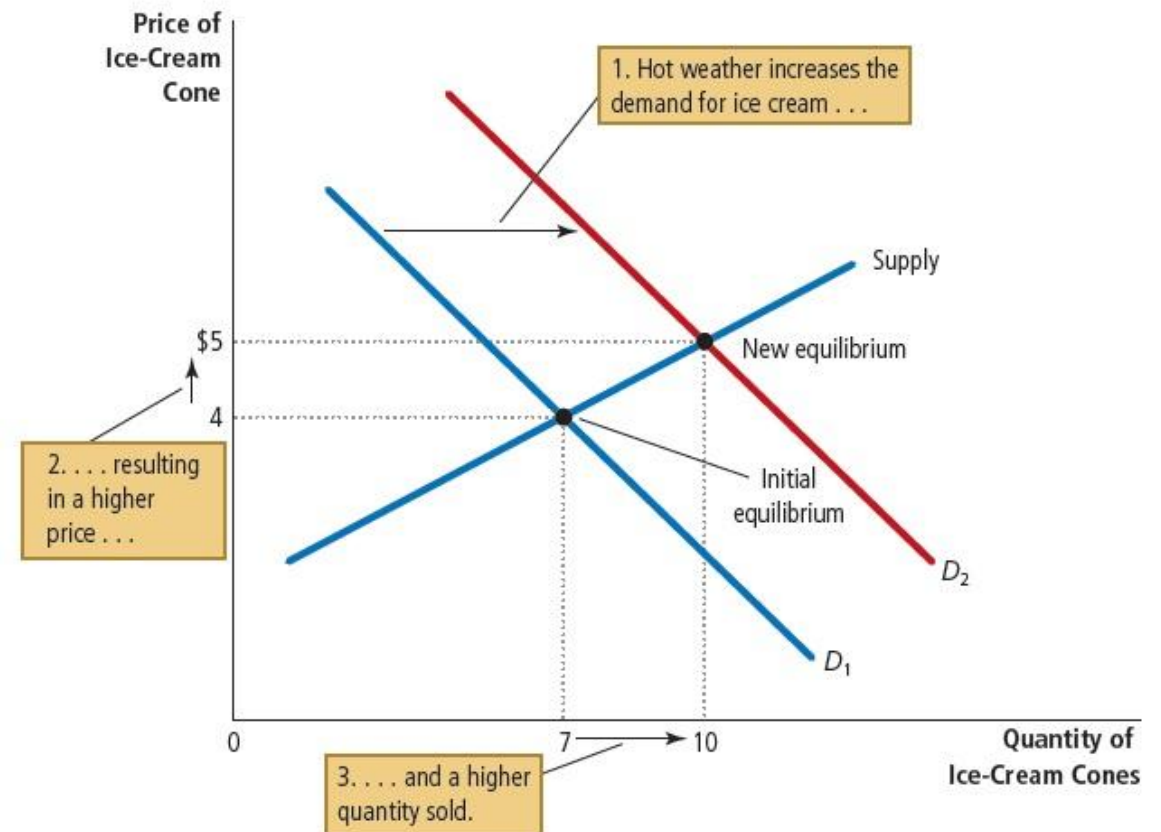


Figure 11 How a Decrease in Supply Affects the Equilibrium

- An event that reduces the quantity supplied at any price shifts the supply curve to the left.
- The equilibrium price rises, and the equilibrium quantity falls.
- Here, an increase in the price of sugar (an input) causes sellers to supply less ice cream.
- The supply curve shifts from S_1 to S_2 , causing the equilibrium price of ice cream to rise from \$4 to \$5 and the equilibrium quantity to fall from 7 to 4 cones.

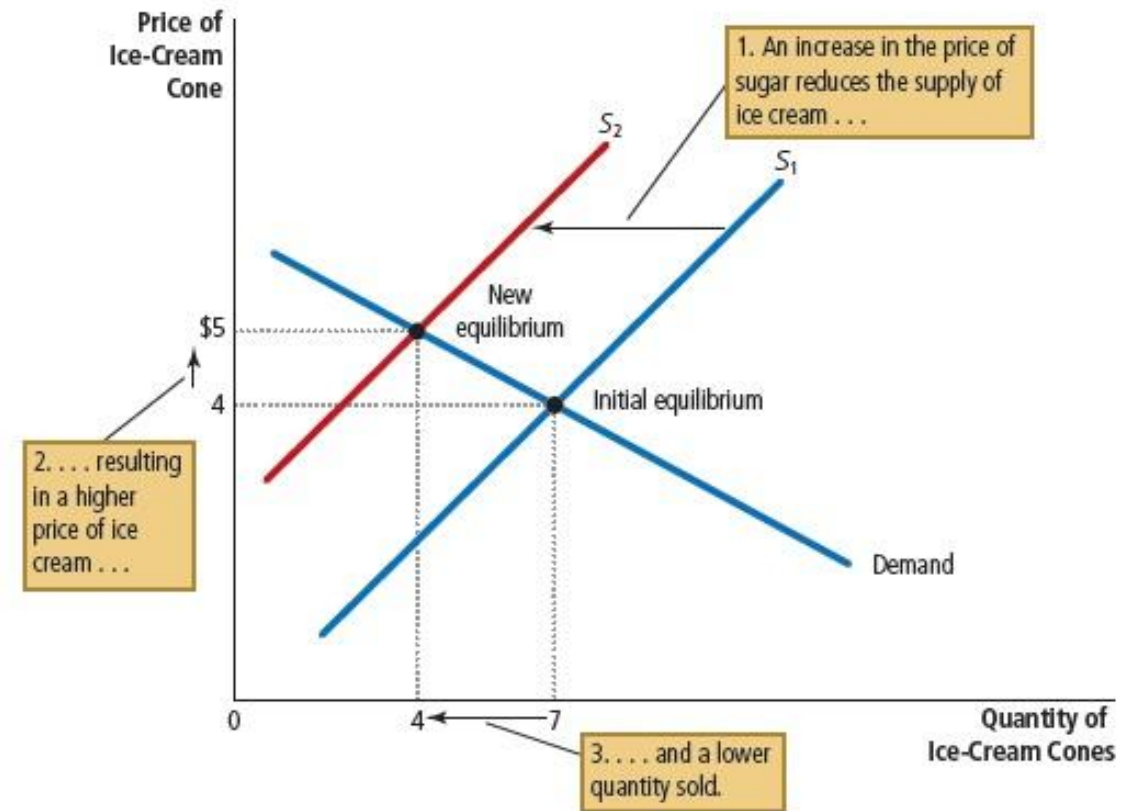
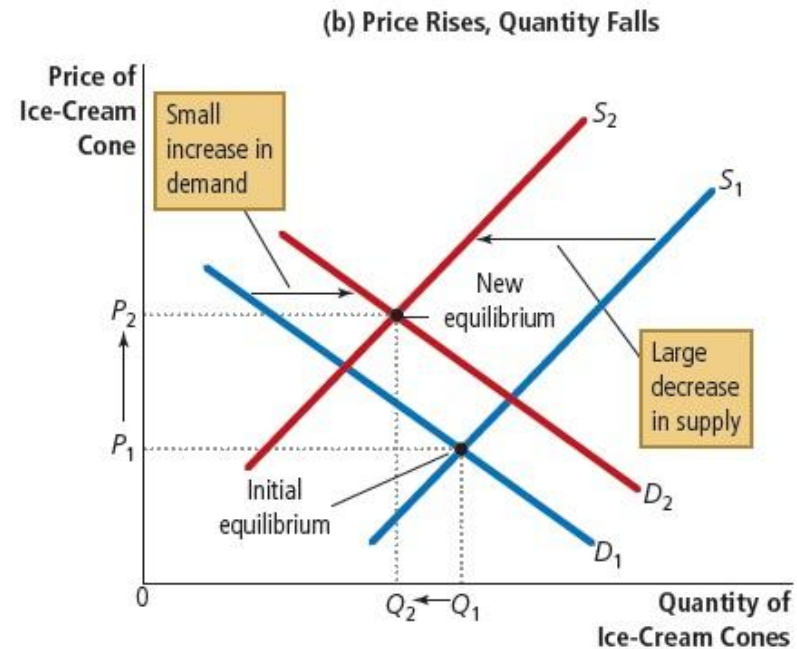
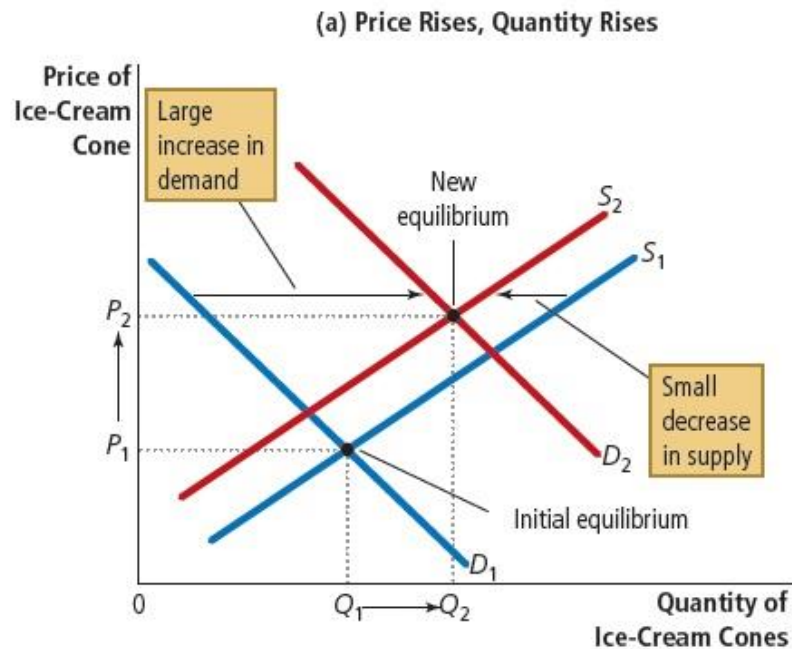


Figure 12 A Shift in Both Supply and Demand

A simultaneous increase in demand and decrease in supply yields two possible outcomes. In panel (a), the equilibrium price rises from P_1 to P_2 , and the equilibrium quantity rises from Q_1 to Q_2 . In panel (b), the equilibrium price again rises from P_1 to P_2 , but the equilibrium quantity falls from Q_1 to Q_2 .



Active Learning 3: Shifts in Supply and Demand

Use the three-step method to analyze the effects of these events on the equilibrium price and quantity of orange juice

- There is a fall in the price of apple juice
- The price of oranges declines because of an abundant orange crop
- Both events occur simultaneously

Active Learning 3: Answers

- There is a fall in the price of apple juice
- The price of oranges declines because of an abundant orange crop
- Both events occur simultaneously
- Steps:
 1. Both curves shift
 2. D shifts left, S shifts right
 3. P falls, effect on Q is ambiguous

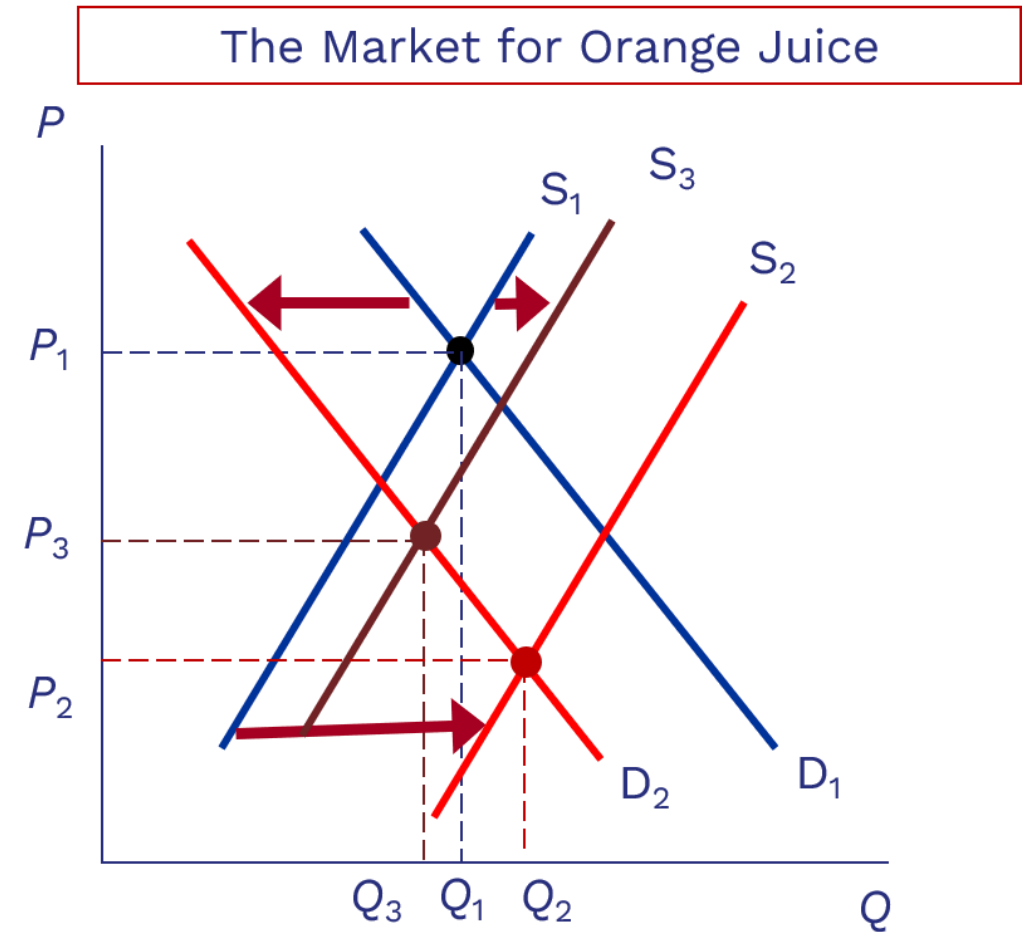


Table 4 What Happens to Price and Quantity When Supply or Demand Shifts?

As a quick quiz, make sure you can explain at least a few of the entries in this table using a supply-and-demand diagram.

	No Change in Supply	An Increase in Supply	A Decrease in Supply
No Change in Demand	P same Q same	P down Q up	P up Q down
An Increase in Demand	P up Q up	P ambiguous Q up	P up Q ambiguous
A Decrease in Demand	P down Q down	P down Q ambiguous	P ambiguous Q down

4.5 Conclusion: How Prices Allocate Resources

- In market economies, prices are the signals that guide decisions and allocate scarce resources
- For every good in the economy, the price ensures that supply and demand are in balance
- The equilibrium price determines how much buyers choose to consume and how much sellers choose to produce

Think-Pair-Share Activity

You are watching a national news broadcast. It is reported that a typhoon is heading for the Washington coast and that it will likely destroy much of this year's apple crop. Your roommate says, "This is not going to affect me, I don't eat apples, I only drink pineapple smoothies."

- A. As an eager economics student, what's your response going to be? Explain.
- B. What other markets will be impacted by the destroyed apple crop? How?